

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira Practical

Practical Task IV

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Q.1 In a college:

- 20% of students know Python.
- 80% of students know Java.
- 90% of Python students get placed.
- 40% of Java students get placed.

A randomly selected student is placed.

Find:

- Probability that the student knows Python.
- Probability that the student knows Java.
- Probability that a placed student knows Python.

Code:

`P_Python = 0.20`

`P_Java = 0.80`

`P_Placed_Python = 0.90`

`P_Placed_Java = 0.40`

`P_Placed = (P_Placed_Python * P_Python) + (P_Placed_Java * P_Java)`

`P_Python_Placed = (P_Placed_Python * P_Python) / P_Placed`

`print("Probability that the student knows Python:")`

`print(P_Python)`

`print("Probability that the student knows Java:")`

`print(P_Java)`

`print("Probability that a placed student knows Python:")`

`print(round(P_Python_Placed, 4))`

```
print("Percentage:", round(P_Python_Placed * 100, 2), "%")
```

Output:

```
Probability that the student knows Python:
0.2
Probability that the student knows Java:
0.8
Probability that a placed student knows Python:
0.36
Percentage: 36.0 %
```

Q.2 In a city:

- 85% of taxis are Green.
- 15% are Blue.
- A witness correctly identifies the taxi color 80% of the time.

A witness claims the taxi involved in an accident was Blue. What is the probability that the taxi was actually Blue?

Code:

```
P_Blue = 0.15
```

```
P_Green = 0.85
```

```
P_Witness_Blue_Blue = 0.80
```

```
P_Witness_Blue_Green = 0.20
```

```
P_Witness_Blue = (P_Witness_Blue_Blue * P_Blue) + (P_Witness_Blue_Green * P_Green)
```

```
P_Blue_Witness = (P_Witness_Blue_Blue * P_Blue) / P_Witness_Blue
```

```
print("Probability that the taxi was actually Blue:")
```

```
print(round(P_Blue_Witness, 4))
```

```
print("Percentage:", round(P_Blue_Witness * 100, 2), "%")
```

Output:

```
Probability that the taxi was actually Blue:
0.4138
Percentage: 41.38 %
```

Q.3 Scenario: Product Defect Detection

- 5% of products are defective.
- 95% of products are not defective.
- The testing machine correctly identifies a defective product 98% of the time.
- The machine incorrectly marks a good product as defective 3% of the time.
- A product is marked as defective. Find the probability that it is actually defective.

Code:

```
P_Defective = 0.05
```

```
P_NotDefective = 0.95
```

```
P_Test_Defective = 0.98
```

```
P_Test_NotDefective = 0.03
```

```
P_Test = (P_Test_Defective * P_Defective) + (P_Test_NotDefective * P_NotDefective)
```

```
P_Defective_Test = (P_Test_Defective * P_Defective) / P_Test
```

```
print("Probability that the product is actually defective:")
```

```
print(round(P_Defective_Test, 4))
```

```
print("Percentage:", round(P_Defective_Test * 100, 2), "%")
```

Output:

```
Probability that the product is actually defective:
0.6323
Percentage: 63.23 %
```